Retail Sales Analytics

Where does the business make money, and where does it lose it?

Final Capstone Project | Week 6: Git, GitHub and Data Analytics

Tools: Python, Pandas, Matplotlib, Seaborn, Power BI, Git and GitHub

GitHub: https://github.com/kkrishna13696-rgb/retail-sales-analytics

1. Project Overview

This project is an end-to-end analysis of four years of US retail orders. I cleaned the raw data, explored it in Python, built charts and an interactive Power BI dashboard, and turned the findings into business recommendations.

Problem statement

Sales grew every year, but not every sale is a good sale. Which categories, regions and discount levels drive profit, and which ones drain it?

2. Dataset Description

Superstore retail orders from January 2015 to December 2018: **9,994 order lines**, 793 customers, 1,862 products and 49 states, with 21 original columns.

Column group	Columns
Order info	Row ID, Order ID, Order Date, Ship Date, Ship Mode
Customer	Customer ID, Customer Name, Segment
Location	Country, City, State, Postal Code, Region
Product	Product ID, Category, Sub-Category, Product Name
Measures	Sales, Quantity, Discount, Profit

3. Data Cleaning Process

Issue found	Fix applied
806 stray rows (People and Returns lookup tables pasted below the orders); 504 of them exact duplicates	Removed. All 9,994 genuine order lines were kept
11 missing postal codes (Burlington, VT)	Filled from the same city and state
Order Date and Ship Date stored as text	Converted to datetime
Range checks: sales > 0, quantity > 0, discount 0 to 1, ship date >= order date	All passed
New features	Year, Month, Ship Days, Profit Margin %, Loss Making flag

Result: 9,994 rows, 27 columns, 0 missing values.

4. Exploratory Data Analysis

Metric	Value
Total sales	$2.30M
Total profit	$286K
Profit margin	12.5%
Average discount	15.6%
Order lines that lose money	18.7%

Key findings

- Sales grew 51% from 2015 to 2018 ($484K to $733K) and profit almost doubled ($49.5K to $93.4K).
- Discount and profit margin have a strong negative correlation of -0.86, the strongest relationship in the data.
- Furniture is 32% of sales but earns only a 2.5% margin. Technology and Office Supplies earn about 17%.
- Tables, Bookcases and Supplies are loss-making. Tables alone lost $17.7K.
- Outliers (IQR method) are mostly big-ticket machines, copiers and heavily discounted items.
- Sales peak every September, November and December.

5. Data Visualizations

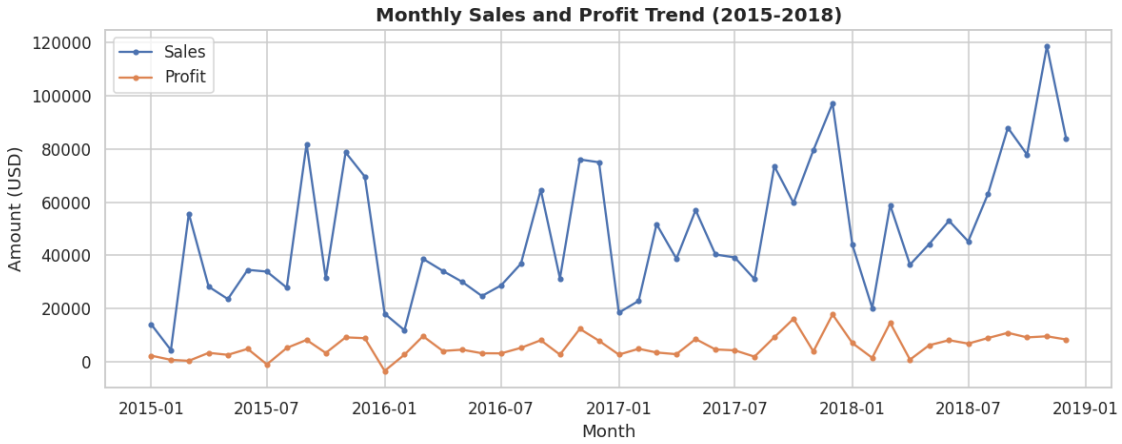


Chart 1: Monthly sales and profit trend

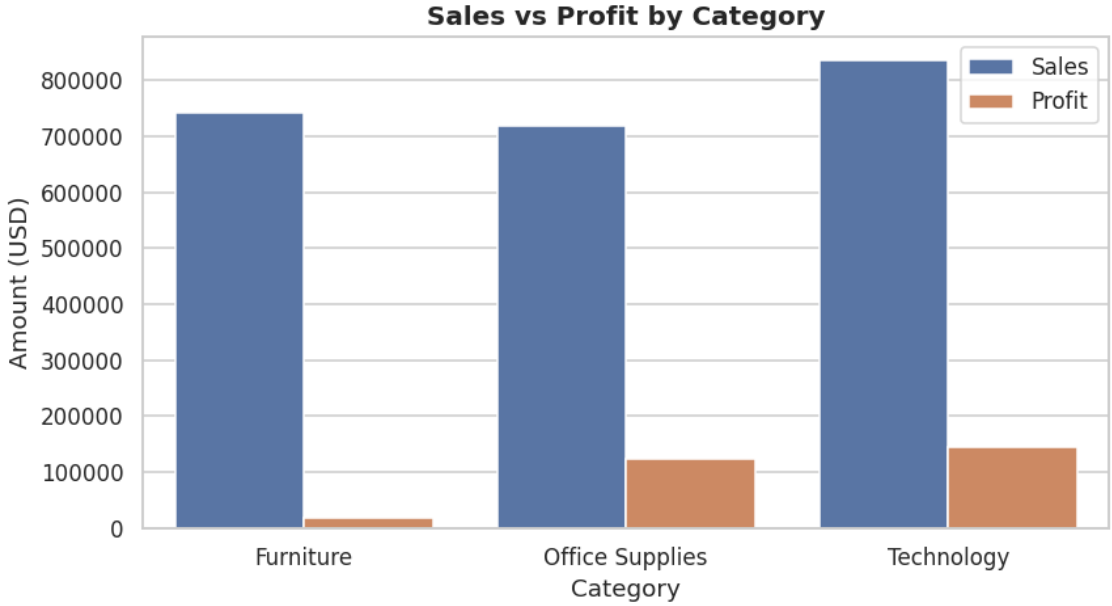


Chart 2: Sales vs profit by category

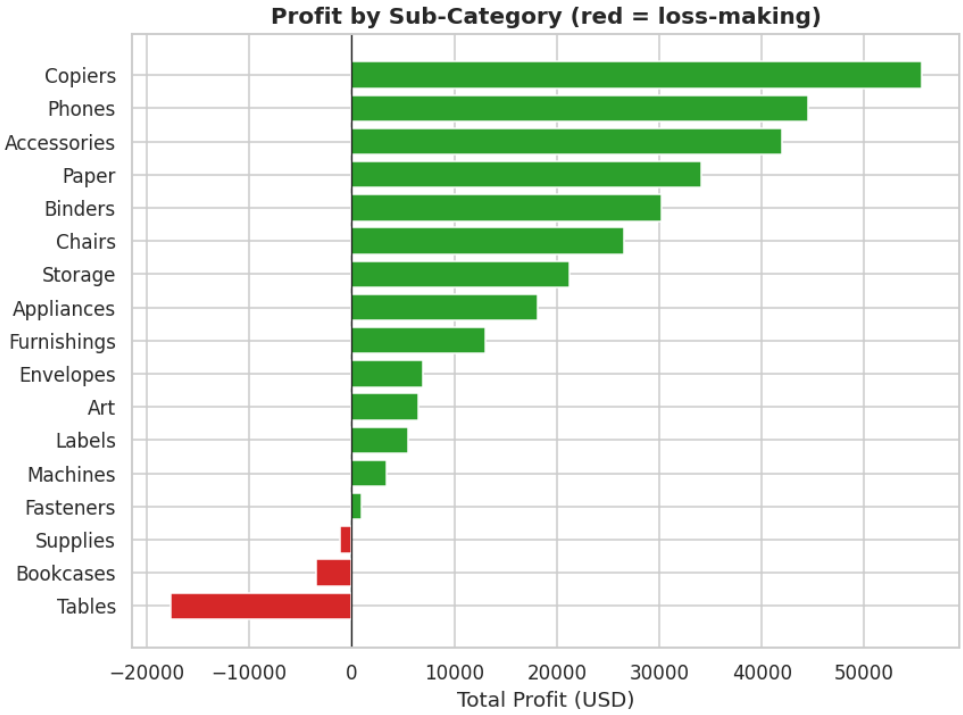


Chart 3: Profit by sub-category

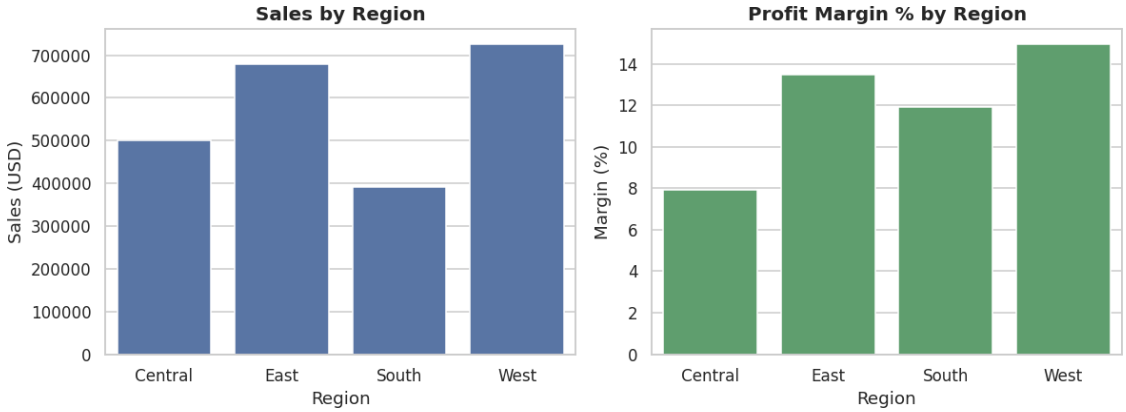


Chart 4: Sales and margin by region

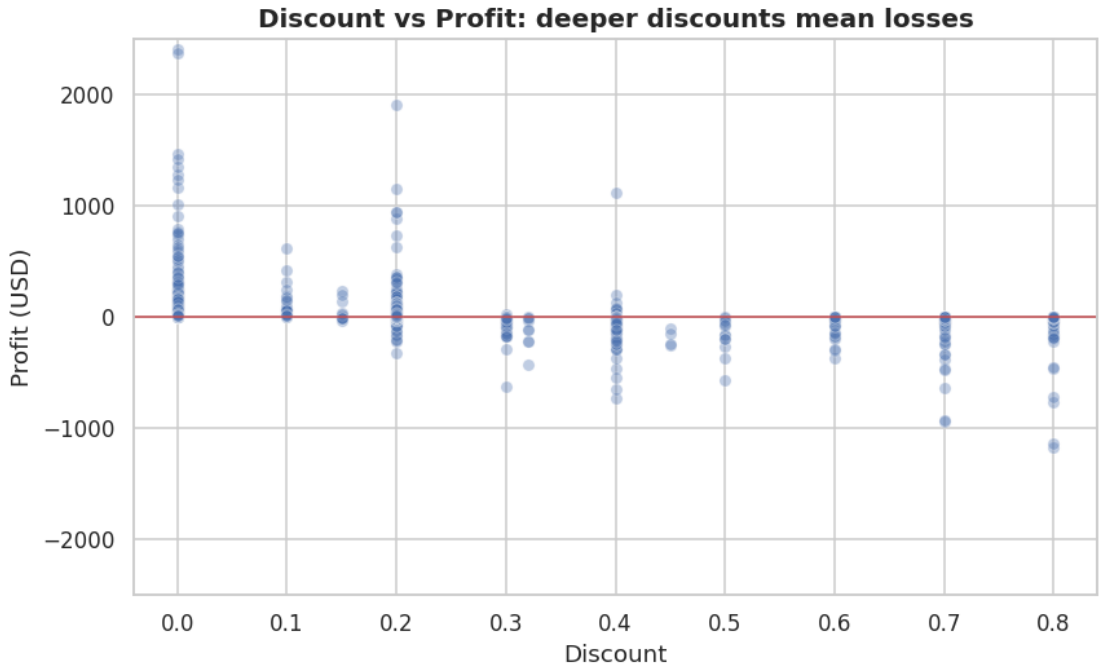


Chart 5: Discount vs profit

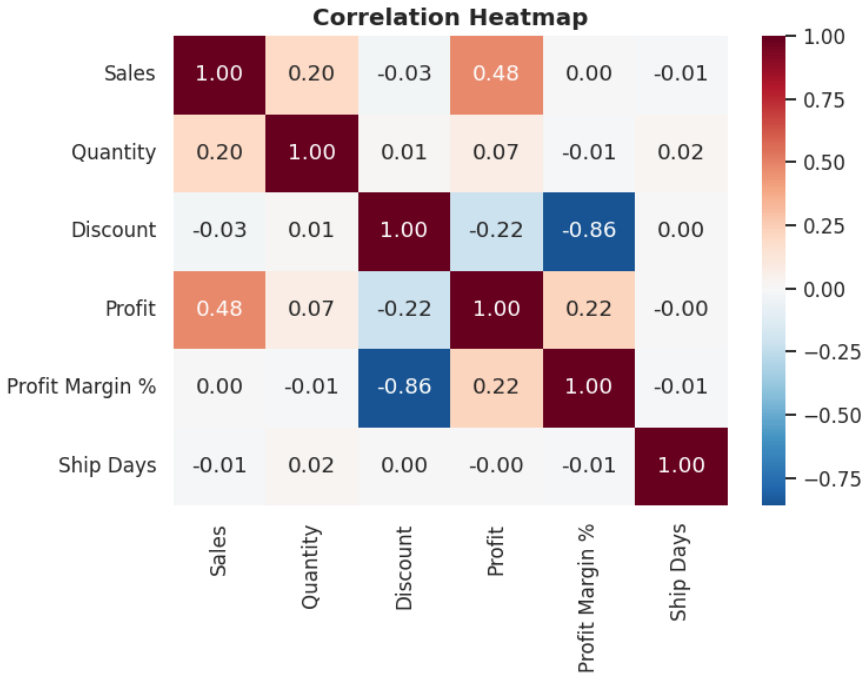


Chart 6: Correlation heatmap

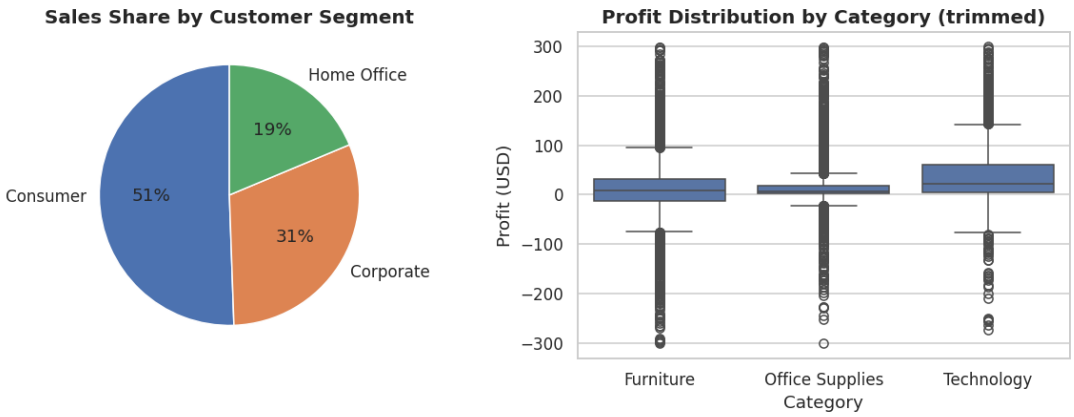


Chart 7: Segment share and profit distribution

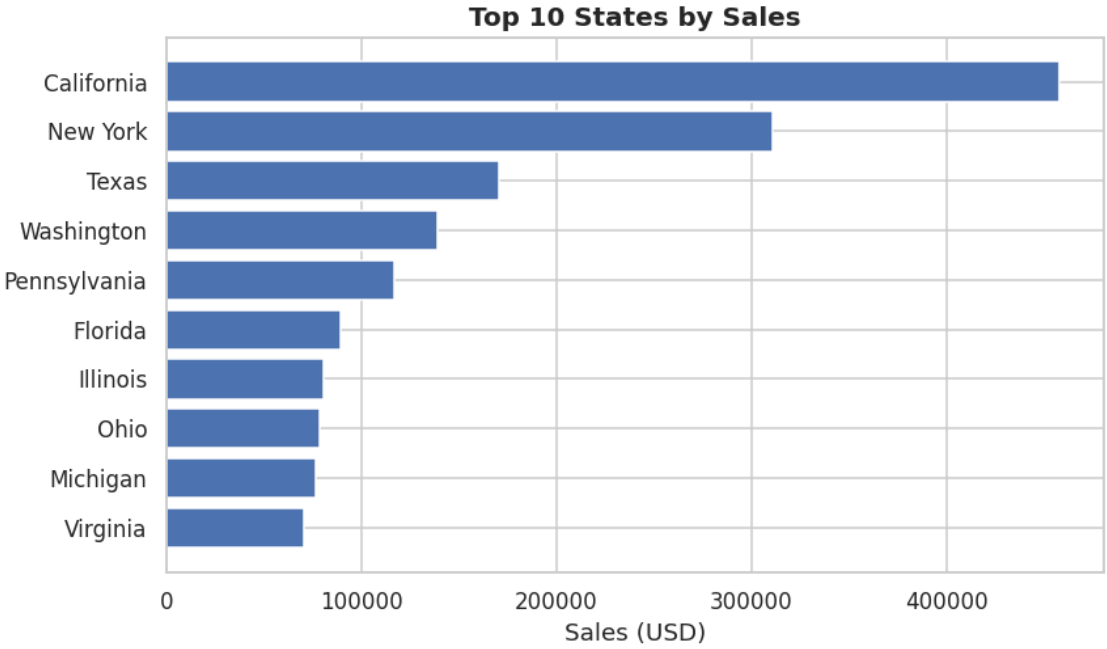
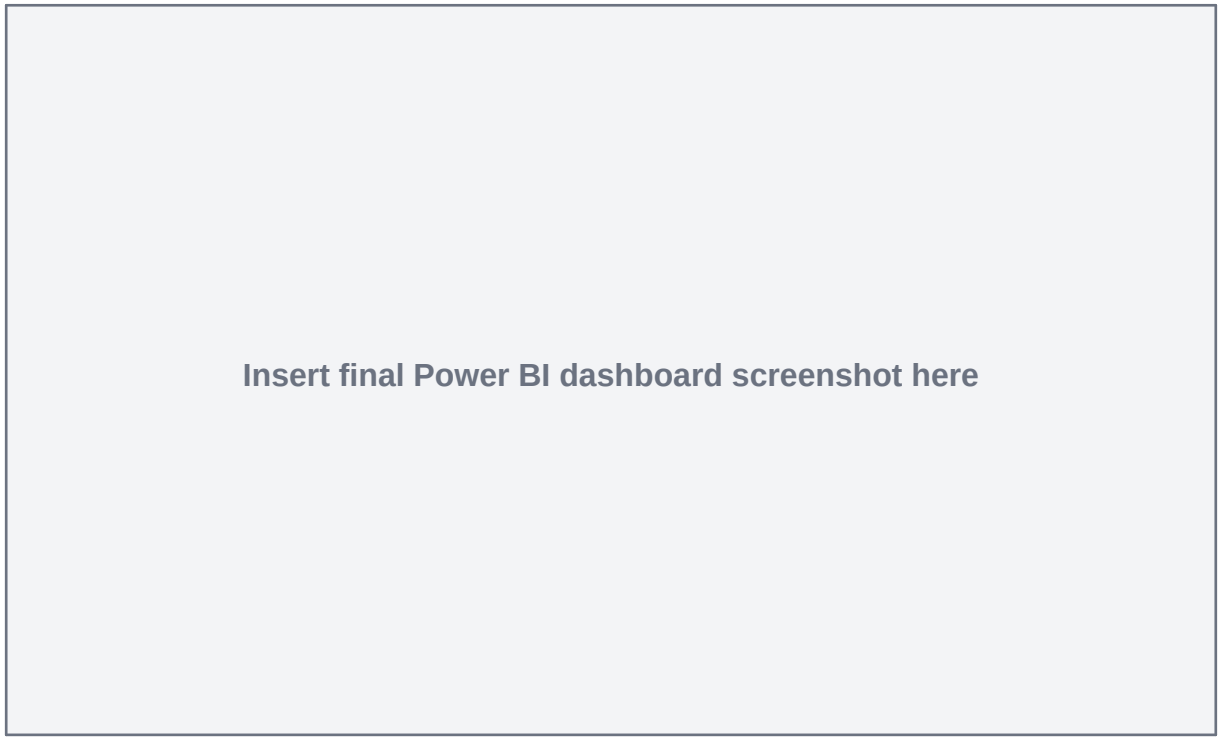


Chart 8: Top 10 states by sales

6. Power BI Dashboard

Interactive dashboard with 4 KPI cards (Total Sales, Total Profit, Margin %, Total Orders), 6 visuals (monthly trend, category, sub-category, segment, state, discount vs margin) and 4 slicers (Year, Region, Category, Segment).



7. Business Insights

- **Healthy growth.** Sales up 51% and profit nearly doubled between 2015 and 2018.
- **Furniture sells a lot but earns almost nothing.** 32% of sales ($742K) but only $18.5K profit (2.5% margin).
- **Tables, Bookcases and Supplies lose money.** Tables lost $17.7K on $207K of sales.
- **Discounts are the biggest profit leak.** No-discount lines earned $321K, while lines discounted 40%+ lost $99.6K across 933 orders.
- **Central region is weakest.** Margin is 7.9% against 14.9% in the West. Texas (-$25.7K), Ohio (-$17.0K), Pennsylvania (-$15.6K) and Illinois (-$12.6K) lose money.
- **Small items are quiet winners.** Paper (43%), Labels (44%), Envelopes (42%) and Copiers (37%, $55.6K profit) have the best margins.
- **Consumers drive volume.** The Consumer segment brings 51% of sales; Corporate and Home Office earn similar margins of 13 to 14%.
- **1 in 5 order lines loses money** (18.7%), clustered in high-discount, big-ticket items.

8. Business Recommendations

- **Cap discounts at 20 to 30% and require approval above that.** Lines discounted 30%+ lost $135K. Even breaking even on them would lift total profit by about 47%.
- **Fix or reprice Tables and Bookcases.** Review supplier cost, shipping and discount policy, or trim the range.
- **Run a Central-region turnaround.** Start with Texas, Ohio, Pennsylvania and Illinois.
- **Push high-margin lines** (Copiers, Accessories, Paper, Labels, Envelopes) through bundles and marketing.

9. Conclusion

Growth is real, but a handful of deep discounts and two weak furniture lines are holding profit back. Fixing discount discipline alone could lift profit by close to half.